MILITARY SPECIFICATION SHEET

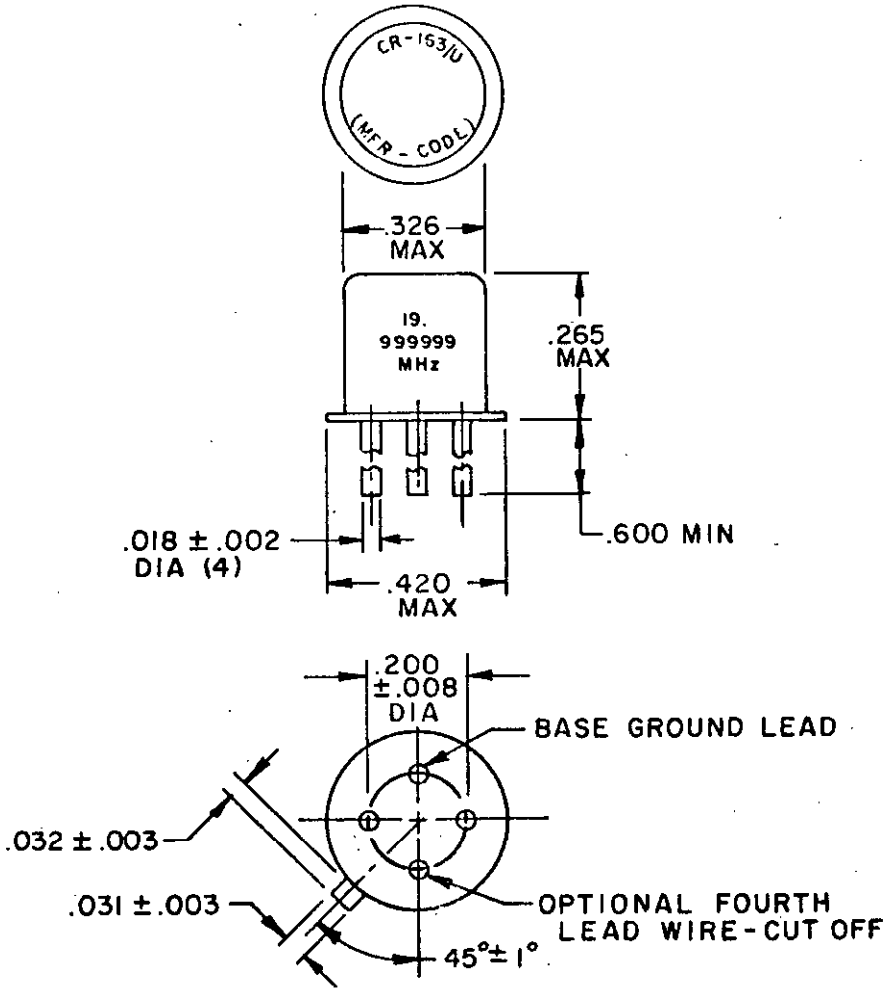
CRYSTAL UNIT, QUARTZ, CR-163/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 7 to 20 MHz; fundamental; noncontrolled; series.

INCHES	MM
.002	.05
.003	.08
.008	.20
.018	.46
.031	.79
.032	.81
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Dimensions and configuration.

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TABLE I. Requirements.

Holder	HC-35/U (see MIL-H-10056/21)
Dimensions and configuration	See figure 1
Frequency range	7 to 20 MHz, inclusive
Frequency tolerance	±20 PPM (-42°C to +90°C) ±40 PPM (-55°C to +105°C) ±5 at 25°C
Equivalent resistance:	
7 to 10 MHz	30 ohms
10+ to 20 MHz	25 ohms
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Operating temperature range	-55° to +105°C, inclusive
Capacitance, shunt	7 pF, maximum
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent-resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent-resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±3 PPM
Equivalent-resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±3 PPM

1/ Method 204 of MIL-STD-202, test condition D.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	7 to 10	13	10	---	±.005 ±.025
TS-683/TSM	10+ to 20	10	--	.07	

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 19
DLA - ES

User activities:
Navy - AS, MC
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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